

ABHISHEK KUMAR GUPTA

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Profile Summary

Robotics and UAV/drone engineering student (B.Tech IT) building **autonomous systems** and **autonomous navigation** for drones in GPS-denied environments. Hands-on with **Raspberry Pi**, **LiDAR**, **computer vision**, **SLAM fundamentals**, **sensor integration**, and **embedded systems** (Arduino, C++, Python).

Education

Rustamji Institute of Technology (RJIT)

2024 – 2028

B.Tech in Information Technology

Gwalior, India

Army Public School, Kunraghat

2023 – 2024

Class XII (CBSE)

Gorakhpur, India

Experience

Robotics Intern | AatonovaZ Technologies Private Limited

08 Sept 2026 – Present

- Working with the **hardware team** on the development of a **drone flight controller (FC)** and related hardware systems.

Jawahar Navodaya Vidyalaya

05 Nov 2025 – 27 Nov 2025

Robotics Trainer

Nalanda, Bihar

- Trained student teams in **Arduino**, **sensors**, and core **robotics fundamentals** through hands-on **embedded systems** sessions.
- Mentored teams for robotics competitions, guiding design and debugging of sensor-driven, competition-ready builds.

Motion Cut

Frontend Web Developer Intern

Remote

- Assisted in frontend development and UI improvements using **HTML**, **CSS**, and **JavaScript**, collaborating with the development team.

Projects

NIDAR AirMouse Project | Raspberry Pi, LiDAR, SLAM, Computer Vision, UAV

2026 – Present

- Developing an autonomous UAV navigation system for **GPS-denied environments** using a **Raspberry Pi** as the onboard compute platform.
- Integrating **LiDAR** for depth perception and a **computer vision** pipeline for object and person detection.
- Exploring **SLAM fundamentals** for environmental mapping and Raspberry Pi-flight-controller communication.

Micromouse Autonomous Robot | C++, PID, VL53L0X, Encoders, TB6612FNG

2025

- Designed a high-speed autonomous maze-solving robot using **PID control** and path-planning algorithms on **Arduino Mega**.
- Integrated **VL53L0X ToF sensors** and encoder feedback with **TB6612FNG** motor drivers for real-time maze navigation.

Autonomous Drone / UAV | Flight Stabilization, Sensors

2024

- Implemented basic **flight stabilization** concepts and **sensor integration** for steady flight, building foundational UAV architecture knowledge.
- Built an embedded, autonomous line-following **utility robot** and a fixed-wing **RC aircraft** using **Embedded C** and **CAD**.

Technical Skills

Programming: C++, Python, Embedded C, Arduino, HTML, CSS, JavaScript, SQL

Embedded & Hardware: Raspberry Pi, Arduino, ESP, Sensors, LiDAR, ToF Sensors, Encoders, Motor Drivers, GPIO, UART, I2C, SPI, PCB Design

Robotics & UAV: UAV Systems, Autonomous Navigation, Sensor Integration & Fusion, SLAM Fundamentals, Computer Vision, Real-Time Control

Tools: Linux, Git/GitHub, Arduino IDE, CAD, 3D Printing

Achievements

2nd Position | Micromouse Competition, IIT BHU

2026

1st Position | Micromouse, IIT Patna Celesta

2025

7th Position | Micromouse, IIT Guwahati

2025

Idea Selected | Smart India Hackathon (SIH)

2024

Certifications

AI/ML Certification, Big Data Analytics, Google Certification, Oracle Cloud Infrastructure, CAD & 3D Printing, Team Robolution Member